Bulk-edge correspondence in topological states of matter

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1. Motivation: topology and correspondences between bulk and boundary theories

Topological states of matter are states of matter which, besides being described by the standard local approaches of condensed matter physics and statistical mechanics (for example Landau theory for critical phenomena), exhibit a whole series of global properties that can be quantified by topological invariants. Different examples of systems exhibiting topological properties, both lattice theories and field theories, share a fundamental property: the spectrum is gapped, i.e. the first excited state is a finite amount $\Delta E > 0$ away from the ground state. Lattice theories with this property are called insulators, while in field theory they more generally go by the name of gapped theories. The reasons why topological notions can come up can be many; here we report two brief examples. The first is the lattice Hamiltonian of the SSH model:

$$H = v \sum_n c_B^\dagger(n) c_A(n) + w \sum_n c_B^\dagger(n) c_A(n+1) + h.c. \tag{1.1}$$

Where v, w are two positive real parameters. Since the problem is invariant under lattice translations, we can rewrite the Hamiltonian as a function of the variable k that labels the Brillouin zone, with $k \in S^1$. The model is characterised by two energy bands:

$$E_{SSH} = \pm\sqrt{(v + w\cos(k))^2 + (w\sin(k))^2}. \tag{1.2}$$

If $v \neq w$, the two energy levels never touch and therefore the system is an insulator. The eigenstates live on the Bloch sphere, and can be written as:

$$|u_\pm(k)\rangle = \frac{1}{\sqrt{2}}(1, e^{\pm i\phi(k)}), \quad \phi(k) = \arctan\left(\frac{\sin(k)}{\cos(k) + \frac{v}{w}}\right). \tag{1.3}$$

This construction is interesting because $\phi(k) : \ S^1 \to S^1$ can be interpreted as a map, and one can study the winding number associated with the map. One finds for example that $\nu = 0$ if $v > w$ and $\nu = 1$ and $v < w$.

The second example is given by the infrared limits of the RG of gapped theories. Once the RG has reached the energy $E < E_1$, with E_1 the energy of the first excited state, there are no more degrees of freedom corresponding to the excitations of the fields. What remains is either a trivial theory or a theory whose only degrees of freedom are topological. This is the situation observed when dealing with Chern-Simons theories, which will be studied in chapter 2.

Some important features of these theories are observed by studying the properties of the boundary of these theories. In general one expects that, having only information on what happens on a boundary of a theory, it is unlikely to deduce anything about what happens instead in the bulk. Surprisingly, in many topological theories one observes genuine correspondences between the bulk and the boundary, in which different quantities of the two theories can have an equivalent formulation. In the case of the SSH model one can show that for every edge there exist ν boundary states (i.e. states that decay exponentially going into the bulk) with zero energy. In the case of 2D topological materials the boundary theory turns out to be chiral, as described in section 2, and it presents various information about the bulk, described at length in section 3.

2. Quantum hall effect and Chern-Simons theories

In this section we will introduce some useful properties concerning the physical description of the quantum hall effect and its formulation with Chern-Simons theories.

2.1 Interaction between particles: FQHE at filling fraction $1/m$

Quantum hall effects can take place in various $2+1$ D physical systems; in this treatment, for simplicity, we will restrict ourselves to the case in which there is a uniform magnetic field orthogonal to the surface. Under the hypothesis of non-interacting electrons completely filling up to the n-th Landau level, subject to a weak electric field, one can show the quantisation of the electrical conductivity, with value:

$$\sigma_{xy} = \frac{e^2}{h} n. \tag{2.1}$$

The next step is to study the problem of interacting particles. In general the problem is difficult; we will concentrate on the case in which the electrons occupy a fraction $1/m$ of the lowest Landau level, where m is an odd natural number. The approach that follows is heuristic: one tries to guess a wave function of the many-particle system without, however, referring to a specific potential between particles (which is however assumed to be central, as for example in the Coulomb case). In general, being in the lowest Landau level, we have a holomorphic function in the variables z_i:

$$\psi(z_1, .., z_n) = f(z_1, ..., z_n) e^{-\sum_i^n |z_i|^2/(4l_B^2)}. \tag{2.2}$$

Since the wave function must satisfy the Pauli exclusion principle, $f(z_i)$ must be antisymmetric under exchanges $z_i \iff z_j$. Moreover, another property it must have is that of having filling fraction equal to m, hence $N_\phi = mN$ occupied states, so that the total angular momentum required is of order Nm. The simplest wave function satisfying this property is the *Laughlin proposal*, which is written:

$$\psi(z_i) = \prod_{i<j} (z_i - z_j)^m e^{\sum_i |z_i|^2/(4l_B^2)}. \tag{2.3}$$

It is easy to see that with $m = 1$ this wave function reproduces the result obtained by applying the Slater determinant technique with the polynomials z^l, obtaining in particular the Vandermonde determinant. Laughlin's ansatz tends to form a "collective sea" of particles, in which there are no excitations. It is however also possible to add excitations in the form of defects, obtaining more complex wave functions. In particular one finds *quasi-holes*:

$$\psi_{M-holes} = \prod_{k=1}^{M} \prod_{l=1}^{N} (z_l - \eta_k) \prod_{i<j} (z_i - z_j)^m e^{\sum_i |z_i|^2/(4l_B^2)}. \tag{2.4}$$

They are called quasi holes because at η_k the probability of finding the particle vanishes. Through approximate techniques with the Berry connection, one finds that these quasi particles have two interesting properties:

1. they have a fractional charge

$$e^* = \frac{e}{m} \tag{2.5}$$

2. they exhibit anyonic statistics, with phase:

$$\alpha = \frac{1}{m} \tag{2.6}$$

Analogously one can build wave functions of quasi particles with charge opposite to the holes (in which I have peaks of the probability), and neutral excitations.

We conclude this list of properties with a last, particularly relevant one. Moving the QH material onto a torus, one verifies that the translation operators (along cycles not homotopic to the trivial loop) obey the statistics:

$$T_1 T_2 = e^{2\pi i/m} T_2 T_1. \tag{2.7}$$

It is interesting that this algebra, analogous to anyonic statistics, forbids the existence of a single vacuum state. Indeed at least m of them must exist, where the smallest representation is given by:

$$T_1|n\rangle = e^{2\pi i n/m}|n\rangle, \quad T_2|n\rangle = |n+1\rangle. \tag{2.8}$$

In general the degeneracy on a manifold of genus g is given by m^g.

2.2 Chern-Simons theory: introduction

In this section we introduce Chern-Simons theory. Before doing so, however, we give some motivation. The main idea is to want to describe a low-energy effective theory that describes the electromagnetic field A_μ, with matter interaction in $2+1$ D. In this dimension, if we restrict ourselves to considering actions quadratic in the field A_μ, we find a single effective theory:

$$S_{cs} = \frac{k}{4\pi} \int \mathrm{d}^3x \; \epsilon^{\mu\nu\rho} A_\mu \partial_\nu A_\rho. \tag{2.9}$$

Let us make some observations on the action just written.

- From the EOM one finds:

$$F_{\mu\nu} = 0 \tag{2.10}$$

Which implies that there are no local degrees of freedom, since locally one can write $A_\mu = \partial_\mu \phi$. It is not certain, however, that it is possible to write the vector potential globally as written above (for example on compact manifolds); in that case it is possible to write non-trivial topological quantities such as holonomies.

- Chern-Simons theory does not depend on the metric adopted. Indeed, in the formalism of differential forms, letting A be a connection one-form, the action can be rewritten as:

$$S_{cs} = \int_{\mathcal{M}_3} A \wedge \mathrm{d}A. \tag{2.11}$$

From the definition of the energy-momentum tensor it follows directly that $T_{\mu\nu} = 0$; considering the 00 component one finds that the Hamiltonian is identically zero. This fact can be interpreted, once again, as a symptom of the absence of dynamical degrees of freedom, as well as a justification of the fact that Chern-Simons theory is used to represent a gapped theory at energies below the first excited level, hence without particle excitations.

- Finally, since this is the effective theory of the vector potential A_μ, we expect it to be still gauge invariant. A rigorous treatment requires the use of fibre bundle theory to describe the vector potential correctly; here, for brevity, only the main ideas will be reported. Indeed, under a transformation $A_\mu \to A_\mu + \partial_\mu \phi$ the action changes in a non-trivial way as:

$$S_{cs} \to S_{cs} + \frac{k}{4\pi} \int \mathrm{d}^3 x \partial_\mu (\epsilon^{\mu\nu\rho} \partial_\nu A_\rho). \tag{2.12}$$

At this point we consider as manifold $\mathbb{S}^2 \times \mathbb{S}^1$, where the sphere is to be thought of as the spatial part and the circle as the temporal one. From the compactness of the sphere we expect flux quantisation (along the same lines as the magnetic monopole).

$$\frac{1}{2\pi} \int_{\mathbb{S}^2} F_{12} = \frac{\hbar}{e} \tag{2.13}$$

The field ϕ is defined on the time circle modulo 2π; for example it is possible to consider $\phi = \frac{2\pi\hbar\tau}{e\beta}$. With this transformation the action transforms as:

$$S_{cs} \to S_{cs} + \frac{2\pi\hbar^2 k}{e^2} \tag{2.14}$$

Since in the path integral the action always appears in an exponential as $e^{iS/\hbar}$, for there to be gauge invariance it is sufficient that the action change by an integer. This leads to a condition on the parameter k, in particular:

$$\frac{\hbar k}{e^2} \in \mathbb{Z}. \tag{2.15}$$

2.3 Chern-Simons theory and quantum hall effect (brief overview)

Let us now see the link between Chern-Simons theories and the quantum Hall effect. A first result is obtained by considering the definition of the current in an effective theory:

$$J^\mu = \frac{\delta S}{\delta A_\mu} \tag{2.16}$$

Using the CS action one obtains:

$$J_i = -\frac{k}{2\pi} \epsilon_{ij} E_j \quad J_0 = \frac{k}{2\pi} B. \tag{2.17}$$

From the first equation one recovers the integer QH effect iff:

$$k = \frac{e^2 \nu}{\hbar}. \tag{2.18}$$

Which is exactly the quantisation condition mentioned earlier. The previous result also tells us that, as we have written it, the CS action is limited to modelling non-interacting electrons. To represent the interaction effectively, a new field a_μ is introduced, which couples to A_μ through the Lagrangian:

$$S_{eff}[A, a] = \frac{e^2}{\hbar} \int \mathrm{d}^3 x \ \frac{1}{2\pi} \epsilon^{\mu\nu\rho} A_\mu \partial_\nu a_\rho - \frac{m}{4\pi} a_\mu \partial_\nu a_\rho + ... \tag{2.19}$$

Note that only a_μ is a dynamical field, while A_μ is to be thought of as a parameter. One can show that with this Lagrangian the conductivity is written as:

$$\sigma_{xy} = \frac{e^2}{2\pi\hbar m} \tag{2.20}$$

Which is the result obtained for the FQHE. The quasi particles are obtained by coupling to the field a_μ a current $\Delta S = \int a_\mu j^\mu$, where j^μ can be either given or dynamical, defined with a further field ϕ. Finally, we note that what we have obtained is valid for a filling fraction of the first Landau level equal to $1/m$. The most general formulation is obtained with the formalism of K matrices, writing:

$$S_K[a^i, A] = \int \mathrm{d}^3x \frac{1}{4\pi} K_{ij} \epsilon^{\mu\nu\rho} a^i_\mu \partial_\nu a^j_\rho + \frac{1}{2\pi} t_i \epsilon^{\mu\nu\rho} A_\mu \partial_\nu a^i_\rho. \tag{2.21}$$

The characteristic results of this theory are:

$$\sigma_{xy} = (K^{-1})^{ij} t_i t_j, \quad (e^*)^i = (K^{-1})^{ij} t_j, \quad \alpha^{ij} = (K^{-1})^{ij}, \quad \deg = |\det(K)|^g. \tag{2.22}$$

3. Boundary theory and correspondence

Once we have defined the main properties of Chern-Simons theories and their interpretation in the context of the QH effect, before introducing the bulk-edge correspondence it is necessary to talk about a boundary theory. There can be two notions of "boundary". The first, stricter one refers to the boundary of a 3-dimensional manifold on which the Chern-Simons theory is defined. The second, more general one simply refers to a theory defined on a 2-dimensional manifold (spacetime), without necessarily referring to a bulk of which it is the boundary. In the following treatment we will refer mainly to the strict meaning, although it is possible to talk about correspondence in the second case as well.

3.1 Gauge breaking and definition of the boundary theory

As we have seen, Chern-Simons theory is an example of a gauge theory in the case in which spacetime has no boundary. When we consider a manifold with boundary we lose part of the arbitrariness of the values the field can take, because of the presence of boundary conditions. Consider as an example the manifold $\mathcal{M} = \{y \in \mathbb{R}^2, y \geq 0\}$, whose boundary is the line $y = 0$. Starting from the Chern Simons theory:

$$S_{CS} = \frac{m}{4\pi} \int_{\mathcal{M}} \mathrm{d}^3x \epsilon^{\mu\nu\rho} a_\mu \partial_\nu a_\rho. \tag{3.1}$$

The variation is written:

$$\delta S = \frac{m}{4\pi} \int_{\mathcal{M}} \mathrm{d}^3x \ \epsilon^{\mu\nu\rho} \Big(\delta a_\mu f_{\nu\rho} - \partial_\mu (a_\nu \delta a_\rho) \Big). \tag{3.2}$$

In order to obtain the same EOM as the theory formulated without boundary, it is necessary to eliminate the contribution of the total derivative. This is possible by imposing the boundary conditions:

$$\alpha a_t + \beta a_x = 0, \quad \forall y = 0. \tag{3.3}$$

Where v will be seen later to have the meaning of a velocity. In particular we choose:

$$a_t - v a_x = 0, \quad \forall y = 0. \tag{3.4}$$

We therefore see that this condition limits the invariance under the $U(1)$ transformation $A_\mu \to A_\mu + \partial_\mu \phi$. After a transformation of this type we have:

$$S_{CS} \to S_{CS} + \frac{m}{4\pi} \int_{y=0} \mathrm{d}x\mathrm{d}t \ \omega(\partial_t a_x - \partial_x a_t). \tag{3.5}$$

To respect the boundary conditions we fix the gauge by imposing:

$$a_t - v a_x = 0 \quad \forall y \geq 0. \tag{3.6}$$

It is convenient to analyse the problem by changing coordinates, introducing the transformations:

$$t' = t, \quad x' = x + vt, \quad y' = y, \quad a'_{t'} = a_t - v a_x, \quad a'_{x'} = a_x, \quad a'_{y'} = a_y. \tag{3.7}$$

Where the transformations on the fields are such as to make the transformation contravariant and to leave the action invariant. In these new coordinates the gauge-fixing condition reads:

$$a'_{t'} = 0. \tag{3.8}$$

Which leads to the EOM:

$$f'_{xy} = 0 \to a'_i = \partial_i \phi. \tag{3.9}$$

If we were on a manifold without boundary, these conditions would say that our field is pure gauge. In reality this statement must be revised in the presence of a boundary. This statement is made concrete by writing the action with the previous EOM substituted into it and checking that it is not trivial.

$$S_{CS} = \frac{m}{4\pi}\int \mathrm{d}^3x \epsilon^{ij} a'_{i'}\partial_t a'_{j'} = \frac{m}{4\pi}\int \mathrm{d}^3 \partial_{x'}\phi\partial_{t'}\partial_{y'}\phi - \partial_{y'}\phi\partial_{t'}\partial_{x'}\phi = \\ \frac{m}{4\pi}\int_{y=0} \mathrm{d}^2x\partial_{t'}\phi\partial_{x'}\phi = \frac{m}{4\pi}\int \mathrm{d}^2x\partial_t\phi\partial_x\phi - v(\partial_x\phi)^2.$$

The action we find in 2D is called the *floreanini-jackiw action*. The new EOM are written:

$$\partial_t\partial_x\phi - v\partial_x^2\phi = 0. \tag{3.10}$$

The physical meaning of this equation becomes clear by defining:

$$\rho(x) = \frac{1}{2\pi}\partial_x\phi(x). \tag{3.11}$$

Substituting:

$$\partial_t\rho - v\partial_x\rho = 0. \tag{3.12}$$

Which is the equation of a chiral wave of velocity v, $\rho = \rho(x + vt)$. This function can be interpreted as a charge moving on the boundary of the theory. The computations can be done starting from

$$S_J = \int \mathrm{d}^3x\ A_\mu J^\mu = \frac{1}{2\pi}\int \mathrm{d}^3x\ \epsilon^{\mu\nu\rho}A_\mu\partial_\nu a_\rho. \tag{3.13}$$

Since A_μ must remain a $U(1)$ gauge field, the last action takes the form:

$$S_J = \frac{1}{2\pi}\int \mathrm{d}^3x\ \epsilon^{\mu\nu\rho}a_\mu\partial_\nu A_\rho. \tag{3.14}$$

Fixing $A_y = 0$ one recovers:

$$S_J = \int \mathrm{d}y\mathrm{d}t A_t\rho + A_x(-v\rho). \tag{3.15}$$

The last equation shows that associated with the gauge field there are a current density ρ and a current $-v\rho$. For technical reasons the field ϕ is defined modulo 2π. From this fact one can derive the quantisation of the charge, when the theory is defined on the disk, from:

$$Q = \int_0^L \mathrm{d}\sigma\rho = \frac{1}{2\pi}\int_0^L \mathrm{d}\sigma\partial_\sigma\phi = \frac{1}{2\pi}(\phi(L) - \phi(0)). \tag{3.16}$$

3.2 Chiral boson

In this section we forget the three-dimensional origin and focus on the chiral theory defined in the previous section, on $\mathbb{S}^1 \times \mathbb{R}$, with the compact variable $\sigma \in (0, L)$.

$$S_{CS} = \frac{m}{4\pi} \int_{\mathbb{S}^1 \times \mathbb{R}} \mathrm{d}\sigma \mathrm{d}t \; \partial_t \phi \partial_\sigma \phi - v(\partial_\sigma \phi)^2. \tag{3.17}$$

To quantise the theory it is convenient to pass to Fourier transforms:

$$\phi(\sigma, t) = \frac{1}{\sqrt{L}} \sum_{n=-\infty}^{n=+\infty} \phi_n(t) e^{2\pi i n \sigma / L}, \quad \rho(\sigma, t) = \frac{1}{\sqrt{L}} \sum_{n=-\infty}^{n=+\infty} \rho_n(t) e^{2\pi i n \sigma / L} \tag{3.18}$$

With the relation

$$\rho_n = \frac{i k_n}{2\pi} \phi_n, \quad k_n = \frac{2\pi n}{L}. \tag{3.19}$$

From the reality condition $\phi_n^* = \phi_{-n}, \rho_n^* = \rho_{-n}$. Given the definition of the k_n we see that $\rho_0 = 0$. To simplify the problem one could impose the constraint $\phi_0 = 0$. It is worth bearing in mind that in reality this is a zero mode and contains information on the invariance under the translation $\phi \to \phi + \alpha$, which is neglected with this simplification. Rewriting the action in terms of these new variables one obtains:

$$S = \frac{m}{4\pi} \int \mathrm{d}t \sum_{n=-\infty}^{n=+\infty} (i k_{-n} \dot{\phi_n} \phi_{-n} + v k_n k_{-n} \phi_n \phi_{-n}) = -\frac{m}{2\pi} \int \mathrm{d}t \sum_{n=0}^{\infty} (i k_n \dot{\phi_n} \phi_{-n} + v k_n^2 \phi_n \phi_{-n}) \tag{3.20}$$

From this expression it is clear that if the $\phi_{n>0}$ play the role of coordinates, the $\phi_{n<0}$ act instead as momenta. in particular the following commutation relations hold:

$$[\phi_n, \phi_{n'}] = \frac{2\pi}{m} \frac{1}{k_n} \delta_{n+n'} \quad [\rho_n, \phi_{n'}] = \frac{i}{m} \delta_{n+n'} \quad [\rho_n, \rho_{n'}] = \frac{k_n}{2\pi m} \delta_{n+n'}. \tag{3.21}$$

Which in the original variables translates into:

$$[\phi(\sigma), \phi(\sigma')] = \frac{\pi i}{m} \mathrm{sign}(\sigma - \sigma') \quad [\rho(\sigma), \phi(\sigma')] = \frac{i}{m} \delta(\sigma - \sigma') \quad [\rho(\sigma), \rho(\sigma')] = -\frac{i}{2\pi m} \partial_\sigma \delta(\sigma - \sigma'). \tag{3.22}$$

Let us now write the Hamiltonian, given by:

$$H = \frac{mv}{2\pi} \sum_{n=0}^{\infty} k_n^2 \phi_n \phi_{-n} = 2\pi m v \sum_{n=0}^{\infty} \rho_n \rho_{-n}. \tag{3.23}$$

Which lead to the equations of motion:

$$\dot{\rho}_n = i[H, \rho_n] = i v k_n \rho_n. \tag{3.24}$$

Which are equivalent to the solutions of infinitely many harmonic oscillators. We can therefore build the spectrum as:

$$\rho_{-n}|0\rangle = 0, \quad |\psi\rangle = \sum_{n=1}^{\infty} \rho_n^{d_n} |0\rangle, \quad H|\psi\rangle = \frac{2\pi v}{L} \sum_{n=1}^{\infty} n d_n |\psi\rangle = v \sum_{n=1}^{\infty} k_n d_n |\psi\rangle. \tag{3.25}$$

Once the basic structure of the chiral boson theory has been described, we want to find fields that represent "charge carriers", such as the electron and the pseudoparticles. We see that one of the operators that plays this role is the *vertex operator*

$$\Psi :=: e^{im\phi} : \tag{3.26}$$

Where the colons indicate a normal ordering in which the ϕ_{-n} are to be placed on the right. The fact that they are charge-carrying operators is suggested by the commutator with ρ.

$$[\rho(\sigma), \Psi^\dagger(\sigma')] = \Psi^\dagger(\sigma')\delta(\sigma - \sigma'), \quad [\rho(\sigma), \Psi(\sigma')] = -\Psi(\sigma')\delta(\sigma - \sigma') \tag{3.27}$$

These relations show how these fields carry a charge ± 1. We can also study the statistics of these operators, which from BCH is seen to satisfy:

$$\Psi(\sigma)\Psi(\sigma') = e^{-m^2[\phi(\sigma),\phi(\sigma')]}\Psi(\sigma')\Psi(\sigma). \tag{3.28}$$

which leads to $[\Psi, \Psi] = 0$ (bosons) if m is even and $\{\Psi, \Psi\} = 0$ (fermions) if m is odd. Another operator we could study is given by

$$\Psi_{qm} :=: e^{i\phi} : . \tag{3.29}$$

With the commutators with ρ

$$[\rho(\sigma), \Psi^\dagger_{qh}(\sigma')] = \frac{1}{m}\Psi^\dagger_{qh}(\sigma')\delta(\sigma - \sigma'), \quad [\rho(\sigma), \Psi_{qh}(\sigma')] = -\frac{1}{m}\Psi_{qh}(\sigma')\delta(\sigma - \sigma') \tag{3.30}$$

And the statistics:

$$\Psi_{qh}(\sigma)\Psi_{qh}(\sigma') = e^{\pm i\pi/m}\Psi_{qh}(\sigma')\Psi_{qh}(\sigma). \tag{3.31}$$

Suggesting that the field Ψ_{qh} corresponds to the anyonic quasi particles typical of a QH material. What we have shown is that the boundary theory encodes the particle content, and the respective statistics, of the theory that we expect to inhabit the bulk.

3.3 Propagator and experimental consequences

In this section we begin to compute the correlation functions of the operators analysed above, starting from their propagators. Starting from the field ϕ, knowing that it corresponds to a free chiral field, we can compute its propagator as follows: consider the free classic theory:

$$S = \frac{m}{8\pi}\int \mathrm{d}^2x \ \partial_i\varphi\partial^i\varphi. \tag{3.32}$$

The propagator is known to take the form:

$$\langle\varphi(x,t)\varphi(0,0)\rangle = -\frac{1}{m}\log\left(v^2t^2 - x^2\right). \tag{3.33}$$

Decomposing the field $\varphi = \varphi_L + \varphi_R$ we can claim that:

$$\langle\varphi_L(x,t)\varphi_L(0,0)\rangle = -\frac{1}{m}\log(vt + x) + C. \tag{3.34}$$

This is what happens with our field ϕ. Note that the logarithmic dependence suggests that the field ϕ is not really physical. Differentiating twice one verifies that for ρ:

$$\langle \rho(x+vt), \rho(0) \rangle = -\frac{1}{(2\pi)^2 m} \frac{1}{(x+vt)^2}. \tag{3.35}$$

Now we want to compute the propagator of the electronic operators:

$$G_F(x,t) := \langle \Psi^\dagger(x,t)\Psi(0,0) \rangle. \tag{3.36}$$

To do this, it is necessary to open a small parenthesis on the properties of the coherent states of the harmonic oscillator, reported in the appendix. we find:

$$G_F(x,t) = \exp\left(m^2 \langle \phi(x,t)\phi(0,0) \rangle\right) = \frac{1}{(x+vt)^m}. \tag{3.37}$$

The importance of this result comes from the fact that we can derive the anomalous dimension of the field Ψ,

$$\dim(\Psi) = m/2. \tag{3.38}$$

This fact gives us information on the corresponding RG flow. As an example of application, let us consider an interaction of *tunnel junction* type, in which a QH material is separated from a wire by a thin insulator. The two materials interact by tunnelling, exchanging electrons. A possible effective theory describes the phenomenon at the point $x = 0$ as:

$$S = \tau \int \mathrm{d}t \; e^{im\phi(0,t)} \Psi_e^\dagger(0,t) + h.c. \tag{3.39}$$

Where Ψ_e is a classic electronic operator in the conductor. By dimensional analysis we find that $\dim(e^{im\phi}) = m/2$, $\dim(\Psi_e) = 1/2$ and therefore $\dim(\tau) = \frac{1-m}{2}$. This analysis tells us that for $m > 1$ the tunnelling modelled here is an irrelevant process at low energies, and that therefore it can be treated using perturbation theory. Using Fermi's golden rule $w = \frac{2\pi}{\hbar}|V_{fi}|^2 \rho(E_f)$ one sees that $G \propto \tau^2$, and since $\dim(G) = 0$ we obtain:

$$G \sim \tau^2 T^{m-1}. \tag{3.40}$$

Or at zero temperature, in the presence of a potential:

$$I \sim \tau^2 V^m. \tag{3.41}$$

This configuration has been tested for $m = 3$, producing confirmations of what was expected theoretically! In the situation in which both temperature and potential are non-zero, one can show that the first contributions are:

$$I \propto T^m (x + x^m), \quad x = \frac{eV}{2\pi k_B T}. \tag{3.42}$$

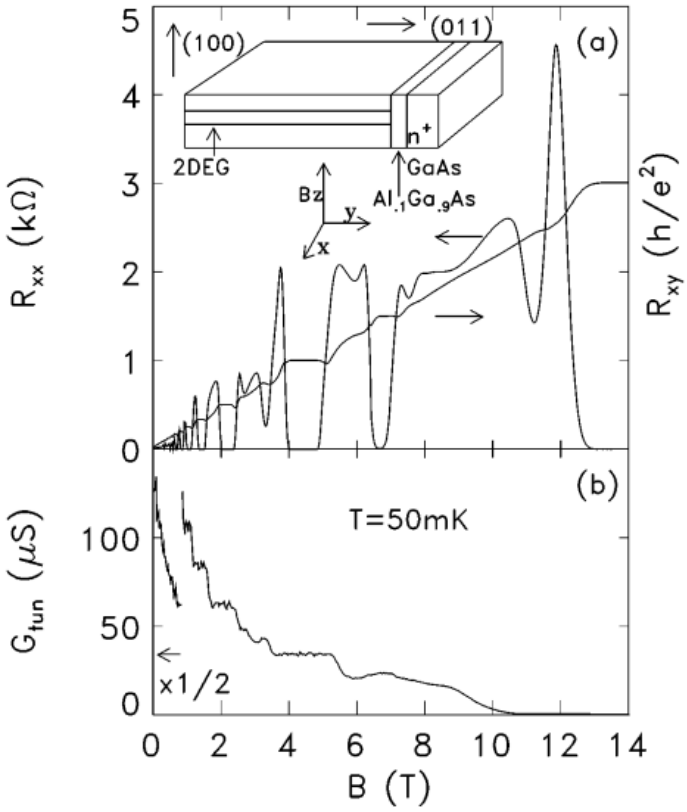


FIG. 1. Magnetic field traces of (a) longitudinal resistance (R_{xx}) and Hall resistance (R_{xy}), (b) tunneling conductance ($G_{\rm tun}$), at low bias for sample 1. The temperature is 50 mK.

In figure 3.3 and table 3.3 the configuration and the experimental parameters are shown. For the experiment to be feasible it is important that the edge of the QH material decays with a decay length smaller than the magnetic one. In the case in question the decay length is about 3 nm, against $l_B = 7,8\ nm$.

TABLE I. Sample parameters.

Sample	n (10^{11} cm^{-2})	μ (cm^2/V s)	E_F (meV)	E_0 (meV)	QW (Å)	Barrier (Å)	ν	B (T)	Δ (μeV)	$\alpha(I$-$V)$	$\alpha - 1[G(T)]$
1	1.08	2.9×10^6	3.9	23	250	90	1/3	13.4	450	2.70	1.75
2	0.87	1.4×10^6	3.1	23	250	90	1/3	10.8	320	2.65	1.50
3	1.16	3.3×10^6	4.1	23	250	225	1	4.8	1450	1.20	0.20
4	2.06	1.6×10^6	7.1	31	250	125	1	8.5	2600	1.14	0.14
5	1.94	1.5×10^6	6.9	31	250	90	2/3	12.0	220	1.20	0.20
6	1.80	2.0×10^6	6.4	30	250	90	2/3	11.1	140	1.42	0.42

3.4 Bulk-boundary correspondence: wave function and correlators on the boundary

We conclude by giving a concrete example of the bulk-boundary correspondence. For simplicity, in this section we set $v = 1$, and we pass to the complex plane through the coordinates:

$$w = \frac{2\pi}{L}\sigma + it, \quad \bar{w} = \frac{2\pi}{L}\sigma - it \tag{3.43}$$

Then we move from the cylinder to the plane through the transformation:

$$z = e^{-iw}, \quad \bar{z} = e^{i\bar{w}}. \tag{3.44}$$

The analysis of the previous sections tells us that the chirality of the boson translates into holomorphicity $\phi(z, \bar{z}) = \phi(z)$. While the correlation function takes the value:

$$\langle \phi(z)\phi(w) \rangle = -\frac{1}{m} \log(z - w) + C. \tag{3.45}$$

Let us now try to evaluate the more general correlation function:

$$\langle : e^{im_1\phi(z_1)} :: e^{im_2\phi(z_2)} : ... : e^{im_n\phi(z_n)} : \rangle = \exp\left(\sum_{i<j} m_i m_j \langle \phi(z_i)\phi(z_j) \rangle \right) \sim \prod_{i<j} (z_i - z_j)^{m_i m_j / m}. \tag{3.46}$$

We can sense that this result is not correct. Indeed, by invariance of the correlators under $\phi \to \phi + \alpha$, from the LHS I recover the equation:

$$e^{i\alpha(m_1 + ... + m_n)} = 1 \iff \sum_i m_i = 0. \tag{3.47}$$

We see that this condition is satisfied for the propagator of Ψ, but in general it imposes a strong constraint, which leads to correcting the RHS with a delta:

$$\langle : e^{im_1\phi(z_1)} :: e^{im_2\phi(z_2)} : ... : e^{im_n\phi(z_n)} : \rangle = \exp\left(\sum_{i<j} m_i m_j \langle \phi(z_i)\phi(z_j) \rangle \right) \sim \prod_{i<j} (z_i - z_j)^{m_i m_j / m} \delta(\sum_i m_i). \tag{3.48}$$

To find something that does not have the delta we consider the propagator:

$$G(z_i, \bar{z}_i) = \langle \Psi(z_1)...\Psi(z_n) \exp\Big(-\rho_0 \int_\gamma \mathrm{d}^2 z' \phi(z') \Big) \rangle. \tag{3.49}$$

Where $\rho_0 = 1/(2\pi l_B^2)$ and γ is a disk containing all the points of the correlation function. The inserted operator can be interpreted as a background charge. Shift invariance this time implies:

$$mN = \rho_0 \int_\gamma \mathrm{d}^2 z = \pi R^2 \rho_0 \tag{3.50}$$

OK if $R = \sqrt{2mN} l_B$ is the disk of the QH material! Carrying out the computation one obtains:

$$G(z_i, \bar{z}_i) \sim \prod_{i<j} (z_i - z_j)^m \exp\left(-\rho_0 \sum_{i=1}^{N} \int_\gamma \mathrm{d}^2 z' \log(z_i - z') \right) \tag{3.51}$$

Which leads to:

$$G(z_i, \bar{z}_i) = \prod_{i<j} (z_i - z_j)^m \exp\Big(-\sum_i |z_i|^2 / 4l_B^2 \Big). \tag{3.52}$$

From this computation we recover Laughlin's wave function! Finally, we observe that it is possible to do better, by adding to the correlator the operators of the quasi particles. The result is:

$$G(z_i, \bar{z}_i, \eta_a, \bar{\eta}_a) = \prod_{a<b}(\eta_a - \eta_b)^{1/m} \prod_{a,i}(z_i - \eta_a) \prod_{i<j}(z_i - z_j)^m \exp\Big(- \sum_i |z_i|^2/4l_B^2 - \sum_a |\eta_a|^2/(4l_B^2)\Big). \tag{3.53}$$

Let us observe the relevance of this correspondence, since the wave functions are defined in the bulk without the need of any boundary, while the quantities we have derived live in a CFT in 2D. The link between these two quantities comes from the topological nature of the bulk theory, since both the boundary of the theory and the wave function are defined on two-dimensional spaces (the spacelike wave function) and topology allows us, informally speaking, to "rotate this surface", so as to give rise to the correspondence shown.

4. Conclusions and possible further developments

Above we have given a single example of the bulk-boundary correspondence, which relates the theory on the boundary of a disk to that of the disk itself. As has been pointed out, however, it is not necessary that the 2D theory be the boundary of the theory whose bulk is in correspondence. An example that will not be explained but only mentioned is the correspondence.

$$\textit{Wavefunction in } \mathbb{R} \times \mathbb{S}^2 \iff \text{Partition function of chiral 2D CFT.} \tag{4.1}$$

Examples of this type recall certain similarities with the *dS-CFT* correspondence, which links precisely the wave functions in the bulk with a partition function on the boundary. Finally, note that the whole treatment has been carried out assuming abelian Chern-Simons, and a potentially interesting thing could be to find how the correspondence changes in the non-abelian case.

5. Bibliography

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A. Single particle in a magnetic field: Landau levels

Let us begin the discussion from the problem of a free particle in a magnetic field. Knowing that this field does no work, the energy is given by:

$$H = \frac{1}{2}mv^2. \tag{A.1}$$

Knowing that the electromagnetic Lagrangian from the point of view of the particle is written as $L = \frac{1}{2}mv^2 - qv \cdot A$, which defines the momentum $P = mv - qA$ and hence $v = (P + qA)/m$. Substituting, one obtains the Hamiltonian for the electron:

$$H = \frac{1}{2m}(p + eA)^2. \tag{A.2}$$

To solve the eigenvalue problem in the case of a uniform magnetic field $(0, 0, B)$ it is necessary to choose a vector potential A. Two possible choices, equivalent to each other, are:

1. Landau gauge:$A = (-By, 0, 0)$ and Hamiltonian:

$$H = \frac{1}{2m}P_y^2 + \frac{1}{2m}(P_x - eBY)^2. \tag{A.3}$$

   In this gauge I have explicit invariance under translation along x; substituting the respective value one obtains the Hamiltonian of a harmonic oscillator:

$$H = \frac{1}{2m}P_y^2 + \frac{1}{2}m\omega_B^2(y - kl_B^2)^2. \tag{A.4}$$

   where:

$$\omega_B = \frac{eB}{m}, \quad l_B = \sqrt{\frac{\hbar}{eB}}. \tag{A.5}$$

   The energy eigenvalues are:

$$E_n = (n + \frac{1}{2})\hbar\omega_B. \tag{A.6}$$

   If the problem is defined on a finite strip with sides L_x and L_y, one easily sees that there is a degeneracy given by the integer part of ϕ/ϕ_0, where $\phi_0 = \frac{2\pi\hbar}{e}$.

2. Symmetric gauge: $A = (-\frac{1}{2}By, \frac{1}{2}Bx, 0)$ and Hamiltonian:

$$\frac{1}{2m}(P_x - eBY/2)^2 + \frac{1}{2m}(P_y + eBX/2)^2. \tag{A.7}$$

   In this case what is explicit is the symmetry with respect to the operator L_z (i.e. $[L_z, H] = 0$). It is possible to build two copies of creation and annihilation operators $a^\dagger, b^\dagger$, for which the Hamiltonian and the momentum are written:

$$H = \hbar\omega_B(a^\dagger a + 1/2), \quad L_z\hbar(a^\dagger a - b^\dagger b), \quad [L_z, b^\dagger] = -\hbar b^\dagger. \tag{A.8}$$

   In this way the energy states depend only on a, while the degeneracy is obtained by applying b. Focusing on the lowest level, the eigenstates

$$|0, m\rangle = Az^m e^{-\frac{|z|^2}{4l_B^2}}, \tag{A.9}$$

   are functions peaked on rings of radius $R_m = \sqrt{2m}l_B$ and $z = x - iy$. From the estimate $\pi R_m^2 < A$, one obtains the same degeneracy as in the Landau gauge.

B. Coherent states of the HO

Given the creation and annihilation operators $a, a^\dagger$, we define a coherent state as:

$$|z\rangle := e^{za^\dagger}|0\rangle. \tag{B.1}$$

Two immediate properties are:

$$[a, e^{za^\dagger}] = ze^{za^\dagger}; \tag{B.2}$$

$$a|z\rangle = z|z\rangle. \tag{B.3}$$

Let $A_i := \alpha_i a + \beta_i a^\dagger$, and given the normal ordering $: e^{A_i} :$, we want to compute $\langle : e^{A_1} :: e^{A_2} : ... : e^{A_n} : \rangle$
From BCH we have that

$$e^{\alpha a} e^{\beta a^\dagger} = e^{\beta a^\dagger} e^{\alpha a} e^{\alpha\beta}, \tag{B.4}$$

which, applied in succession, leads to:

$$\langle : e^{A_1} :: e^{A_2} : ... : e^{A_n} : \rangle = \exp\left(\sum_{i<j} \langle A_i A_j \rangle\right). \tag{B.5}$$